The SAL to Eta-Local Structure Step in Cirac–Perez-Garcia–Schuch–Verstraete 2017 Appendix C.2

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1 Source statement

Appendix C.2 of Cirac–Pérez-Garcia–Schuch–Verstraete proves that a simple MPDO satisfying the strong area law has the commuting nearest-neighbor product form

$$\sigma^{(N)}(\mathcal{K}) \propto \prod_{n=1}^{N} B_{n,n+1},$$

where $B_{n,n+1} \geq 0$ and the translated two-site operators commute [CPGSV16, Appendix C.2, Proposition C.6]. In the local source file this is Proposition `3to4`, `Papers/1606.00608/MPDO-22-12-17-2.tex:1571--1593`.

The proof uses the sector decomposition obtained earlier in Appendix C.2:

$$\tilde{\sigma} = \bigoplus_{k_1,\ldots,k_N} \bigotimes_{n=1}^{N} \eta_{k_n,k_{n+1}},$$

with $\eta_{k_n,k_{n+1}} \geq 0$ acting between the right subspace of site n and the left subspace of site $n+1$. It then defines

$$B_{n,n+1} = \sum_{k_n,k_{n+1}} \mathrm{Id}_{H_{L,n}^{(k_n)}} \otimes \eta_{k_n,k_{n+1}} \otimes \mathrm{Id}_{H_{R,n+1}^{(k_{n+1})}},$$

regarded as the identity away from sites $n, n+1$. With this definition, $\tilde{\sigma} = \prod_n B_{n,n+1}$ and the commutation of the translated B-operators follows from their sector-local action.

2 Current formalization

The formal development currently records this as an eta-local structure for K.[1] It contains the assembled positive translation-invariant two-site bond and its

[1] The formal structure is MPOTensor.EtaLocalStructureData.

realization of the finite-chain MPO. From this structure, the development proves the commuting-form property and the GSNNCH–ZCL consequences. After ZCL is assumed, ZCL itself gives the size-2 fusion witness and the transfer-map fusion formulation of the MPDO RFP condition.

The two scope-restricted declarations are assembly statements.[2] They assume the eta-local structure instead of deriving it from the SAL local coordinates. They also carry the ZCL hypothesis needed for the subsequent RFP consequences. Thus they formalize the final assembly step after the $B_{n,n+1}$ have been constructed, not the source theorem that constructs those bonds from SAL.

3 Missing step

The missing source-faithful theorem must construct the formal eta-local structure MPOTensor.EtaLocalStructureData from the Appendix C.2 local data attached to a simple MPDO satisfying SAL. It must not add ZCL as a hypothesis for this construction. Concretely, it must identify the local $\eta_{k,h}$ family with the original tensor coordinates, define the translation-invariant bond $B_{n,n+1}$ above, prove $B_{n,n+1} \geq 0$, prove the commutation relations, and prove the finite-chain realization

$$\sigma^{(N)}(K) = c_N \prod_{n=1}^{N} B_{n,n+1}$$

for positive scalars c_N.

ZCL enters only after this commuting product form has been obtained, in the later step that derives the GSNNCH–ZCL and RFP conclusions.

This is tracked by issue #823. Until that construction exists, the scope-restricted assembly statements must not be presented as the full formalization of Proposition C.6.

4 Classification

This note records an open gap for the SAL-to-bond construction of MPOTensor.EtaLocalStructureData. The two assembly declarations above are scope restrictions: they formalize only the post-assembly step after the nearest-neighbor bonds have already been constructed.

References

[CPGSV16] J. Ignacio Cirac, David Pérez-García, Norbert Schuch, and Frank Verstraete. Matrix product density operators: Renormalization fixed points and boundary theories. Preprint; published version: Annals of Physics 378 (2017), 100–149, 2016.

[2] These declarations are MPOTensor.SimpleMPDOBlockedRFPData.ofSALZCLAndEtaLocalStructure and MPOTensor.simple_mpdo_rfp_chain_of_sal_zcl_and_etaLocalStructure.